

Bean Academy: A Music Composition Game for Beginners with Vocal Query Transcription

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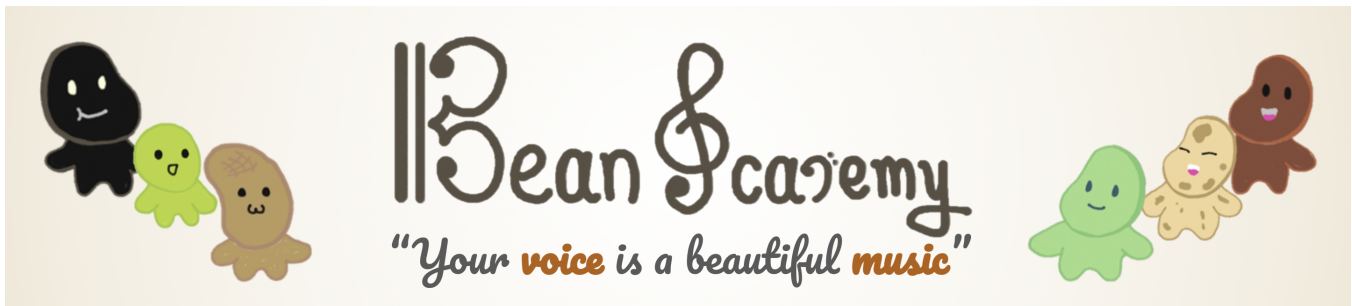


Figure 1: Concept design of *Bean Academy: A Music Composition Game for Beginners with Vocal Query Transcription* (2023)

ABSTRACT

Bean Academy is a music composition game designed for musically-unskilled learners to lower entry barriers to music composition learning such as music theory comprehension, literacy and proficiency in utilizing music composition software. As a solution, *Bean Academy*'s Studio Mode was designed with the adaptation of an auditory-based 'Vocal Query Transcription(VQT)' model to enhance learners' satisfaction and enjoyment towards music composition learning. Through the VQT model, players can experience a simple and efficient music composition process by experiencing their recorded voice input being transcribed into an actual musical piece. Based on our playtest, thematic analysis was conducted in two separate experiment groups. Here, we noticed that although *Bean Academy* does not outperform the current-level Digital Audio Workstation(DAW) in terms of performance or functionality, it can be highly considered as suitable learning material for musically-unskilled learners.

CCS CONCEPTS

• **Applied computing** → **Computer games**; Sound and music computing; E-learning.

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KEYWORDS

Music composition, Vocal Query Transcription, Serious game, Education game

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1 INTRODUCTION

The entry barrier is decreasing in all disciplines as technology becomes increasingly applicable to our daily lives. In particular, technology has developed various aspects of music, especially music making, but compared to this technological enhancement, learning materials and methodologies for understanding is insufficient for those who are new to this field. Digital Audio Workstation(DAW) is a fundamental software used for music and sound generation, yet restrictions in accessing DAW remains as one of the most constrained factors in the current music composition education settings [4, 8, 10]. Vocal Query Transcription(VQT), the method that utilizes the user's input voice to become an actual digital sound, is the solution we have found to lower entry barriers caused by DAW usage [7, 12]. *Bean Academy*, therefore, is an introductory serious game for those who hope to enter the music composition field without being intimidated by the high barrier of entry in learning DAW software. By applying the VQT model to our core game mode 'Studio Mode', we expect to bring more satisfaction and enjoyment towards music composition learning for musically-unskilled players. Based on the iterative design process, we have designed the prototype version of this game using Unity 2022, conducting multiple pilot

build-run tests and playtest workshops for revisions. With the iterative process of improving three game modes, players will be able to engage in music composition in more simple and efficient conditions.

2 RELATED WORKS

2.1 Difficulty in Music Composition Learning

Musical composition is the creative process of generating new music. This pertains to composing melodies, harmonies, rhythms, and the overall structure of a piece of music [2]. Writing one's own music has numerous benefits in terms of health, emotional, and social ways since making music rather than playing existing one necessitates an additional capability: creativity [1]. People's interest in music composition is also increasing. For instance, Soundcloud, a global online music distribution platform where anyone can upload their own songs, has reported that the number of music creators, as well as the revenue of the company, continues to increase consistently, and rapidly [3].

However, musical composition requires high entry barriers: comprehending music theory, and utilizing computer programs, which leads to general people and even musicians giving up on composing music relatively early. Although the Digital Audio Workstation (DAW) has been created to speed up the process of composing or recording and minimize the musical knowledge [8, 10], there are still limitations that the users have to be computer-literate and proficient in dealing with the software itself [4]. Due to these restrictions, we noticed that entry barriers to music composition learning using DAW were too high for our specific target population: musically-unskilled learners who want to compose music without a proper learning environment.

2.2 Vocal Query Transcription

Voice Query Transcription (VQT), is a technique that transcribes or analyzes the input voice to utilize in many ways, especially in musical analysis [7, 12]. VQT technique always begins the process with sampling and encoding the input voice into the calculable figures within the amplitude of the audio. Here, a higher sample rate leads to a higher granularity, which means that it is less likely to lose the information from the original voice input. After that, we can use these values for exploitation in various musical analyses (e.g., onset/offset, pitch, chord detection, etc.) For instance, we can apply the Fast Fourier transform to change the audio from the time into the frequency domain and then find the fundamental frequency for detecting the pitch information [5].

Due to technical limitations, people in the past did not have much access to various applications that use VQT, but in recent years, there are trends in using this technology in many fields, and games are not an exception. For instance, One Hand Clapping is a musical puzzle game, which is played by singing or humming into the microphone to solve musical puzzles and progress through levels [9]. Through the novel utilization of VQT in the gaming context, the majority of game players show high satisfaction and rate it as 'very positive' on the Steam platform [13].

VQT, which has abundant potential and high utilization, seems to have been used in every voice and music-related field, but it was not

easy to find cases used in music composition or education. The fundamental problem we want to address is that musically-unskilled learners' enjoyment and satisfaction towards music composition learning is degraded by DAW, which requires pre-acquired musical comprehension and sufficient literacy and proficiency in software itself [4]. By designing a game applying VQT, it is possible to automatically convert the users' musical flow that comes to their mind into a game screen singing or humming.

3 GAME DESIGN

Based on our prior research, we have noticed that applying the Vocal Query Transcription (VQT) model to our game design will help musically-unskilled players engage in music composition learning since it reduces all huddles accessing the Digital Audio Workstation (DAW) software. *Bean Academy* (the game title) is a music institute where *Professor Beans* (the game characters) teaches basic level and hands-on experience of music creation. The game is constructed with one main mode, Studio Mode, and two supplementary modes, Classroom Mode and Club Mode. In Studio Mode, players can freely create their music clips by simply recording their voices. The auditory model-based VQT algorithm will then utilize the player's recorded data into melodies with different types of instruments, beats, vocals, and post-processed sound mixing. In Classroom Mode and Club Mode, all activities are designed to support and expand their playability in Studio Mode.

3.1 Studio Mode

Studio Mode is the core game activity in *Bean Academy*. Players can feel a magical experience where their voices become promising music through VQT. Based on the players' different musical tastes, they can create their own sound clips in three different sublayers (Record, Melody, and Beat Layer) and manage synthetically in the main layer (Compose Layer).

3.1.1 Record Layer. In Record Layer (see (a) in Figure 2), players can record, edit and decide whether to use the recorded voice in the current music or set it aside in the database so they can exploit it in their later music composition sessions. This layer displays the spectral information of the recorded voice. Players can repeat their recording process until they get a satisfactory result, and if they are satisfied with the current recording, pressing the "DONE" button will enable them to move the result into the Compose Layer (see (d) in Figure 2) for synthetical management.

3.1.2 Melody Layer. In Melody Layer (see (b) in Figure 2), players can experience the VQT model in the best way. After simply recording the player's voices, the model detects the pitch from the input voice and synthesizes it to the type of musical instrument that they have chosen. The prototype version includes a total of 12 different kinds of musical instruments. The model detects the exact pitch and quantizes it into the most reasonable pitch, however, this quantization process can be sometimes unsatisfactory to the players. To cover this problem, players get to drag the transcribed MIDI notes into the desired location for further editing or into the mortar, located on the bottom-left side, to delete specific parts of the transcribed melody. Pressing the "DONE" button will move the result into the Compose Layer for synthetical management.

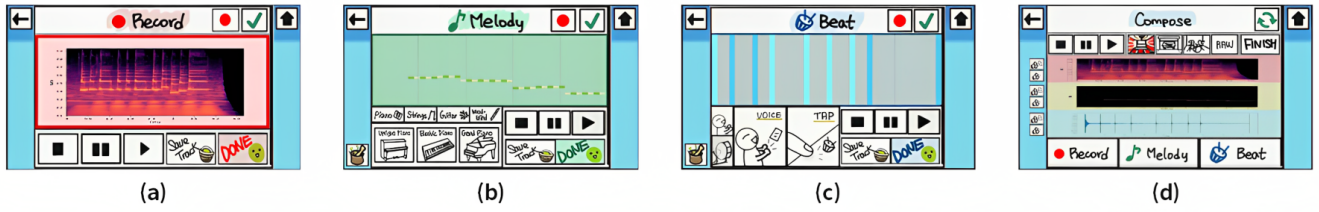


Figure 2: All features of Studio Mode (a. Record Layer, b. Melody Layer, c. Beat Layer, d. Compose Layer)

3.1.3 Beat Layer. In Beat Layer (see (c) in Figure 2), players can utilize their beat input with the percussion-styled synthesis. Players can express their own beats in various methods such as recording voices, clapping, thumping, shouting, etc. The VQT model detects onsets from input music and transcribes them into percussion-style beat music. The prototype version includes the snare and kicks sound of the drum kit. Similar to Melody Layer, this layer provides editing and deleting features so the players can revise the timing of the transcribed beat sound.

3.1.4 Compose Layer. In Compose Layer (see (d) in Figure 2), which is also the main layer of Studio Mode, it shows all the temporarily saved music data (which players have saved by pressing the “DONE” button in each sublayer) and displays them jointly. Players can basically listen to a whole song where all the sublayers have been merged. The VQT model also allows volume adjustments and post-processed sound mixing (e.g., environmental changes whether in the studio, garage or even opera hall). Through going back and forth between the Compose Layer and three sublayers, players can enhance their skills in integrating music as a whole and also determining what is “harmonious and good” music.

3.2 Classroom Mode

During the pilot user study, most of the participants struggled with how to start the composition procedure even if they had the simple tool (Studio Mode using VQT) in front of them. The fundamental reason we have found is that most of them had no prior experience in constructing their music flow in their brain to actual representative form. As a guide to musically-unskilled players, Classroom Mode was designed to support players to learn basic components of music composition by engaging in several mini-games. Furthermore, by adding some exercises that require to use the Studio Mode itself, Classroom Mode will help players to adapt to the composition procedure more quickly.

3.2.1 Instrument Class. In Studio Mode, players can convert their voices into various kinds of instruments and sound effects. To support this feature, Instrument Class allows players to hear the original sounds of each instrument and get information on how and when each instrument is used in the music composition. In the last phase of the class, players can test themselves in the ‘Sound-clip Quiz’ (see (a) in Figure 3) where they are induced to distinguish each instrument’s sound in the mixed record.

3.2.2 Rhythm Class. In the pilot user study, musically-unskilled players had reported difficulties with accurately differentiating the length and tempo of melody lines. To support this difficulty, Rhythm

Class provides a mini-game (see (b) in Figure 3) with detection of how long the players have well-pressed each note, unlike those market-released rhythm games that only focus on the detection of the exact timing of each note. Through these detection changes, musically-unskilled players will feel they are playing various kinds of notes (e.g., half note, quarter note) in the music sheets.

3.2.3 Melody Class. In the pilot user study, musically-unskilled players also reported difficulties with turning the melody flow in their brains into actual voices. This implies that musically-unskilled players normally know fragments of a melody but it is hard for them to put them together into a whole song. The objective of Melody Class is to solve a melody puzzle (see (c) in Figure 3), where players have to arrange separated melody boxes into the correct order.

3.2.4 Final Test. After warming up with different kinds of activities in the prior three classes, the Final Test session is unlocked. In the test, players need to submit their music composition result using Studio Mode, satisfying all the pre-mentioned conditions.

3.3 Club Mode

Club Mode is designed to induce players to expand their music composition results by sharing with other game players. Players can choose which club to join simply on their interest in certain musical genres or instruments. After joining each club, they can listen to other club members’ music or upload their own music to get commentary feedback as well. (see (d) in Figure 3)

3.4 Technical Note

The authors developed the game using Unity 2022 game engine in Visual Studio Code 2022, and visual assets were hand-drawn. Most of the musical analysis in Studio Mode was developed with Python using the librosa library [11]. Unity and the Python server operate through socket communication. The game build supports both Windows and Mac OS.

Because this game analyzes the sound coming into the microphone input device and uses it as a player’s composition activity, that background noise from the recording environment can have unexpected consequences for the user. However, the game has eliminated the noise cancellation function to avoid preventing players from creative activities that try to use not only human voices but also sounds such as clapping and footsteps, and even noise from the surrounding environment. As the subsequent theme analysis shows, the game tends to be played alone in quiet rather than in a public space, so the absence of the noise canceling module would

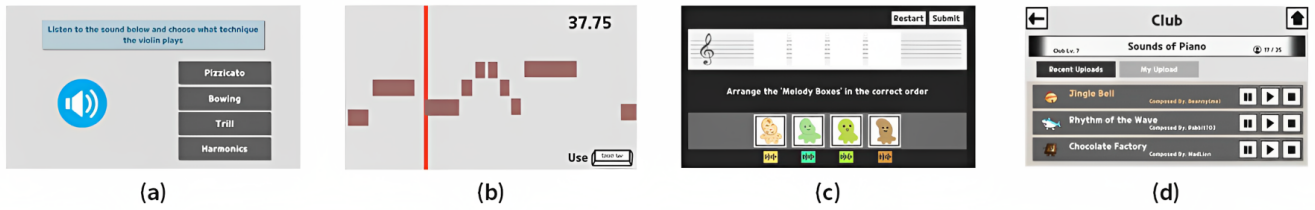


Figure 3: All features of Classroom Mode (a, b, c) and Club Mode (d)

help each user from playing the game according to their desired situation.

4 PLAYTEST

4.1 Test Plan

4.1.1 Participants. 5 players participated in the playtest (3 female, 2 male). All participants were graduate students in their 20s and 30s. Our experiment groups were subdivided into two groups. The first group was the ‘musically-unskilled’ group, which is also our target population. The musically-unskilled group’s participants (1 female, 1 male) had no prior experience in music composition and Digital Audio Workstation(DAW) software usage. The second group was the ‘musically-skilled’ group, which is the opposite of our target population. The reason we recruited this group was to figure out what enhancement our game design needed from the perspective of an expert. Musically-skilled group participants (2 female, 1 male) were skilled at creating melody or rhythm on the spot and had previous DAW experience as amateur musicians and a composition major.

4.1.2 Method. The playtest had three stages. First, a pilot user study using a preliminary survey was completed by participants, in which basic demographic data and a self-evaluation of musical skills were acquired. Next, participants took part in a workshop session in two separate experiment groups that included prototype version gameplay. The Wizard of Oz method was adopted in this stage to acquire more empirical feedback through communicating with the participants while using our game interfaces [6]. Afterward, the discussion session was conducted in an open conversation in a way all participants within an experiment group can freely give an opinion and add to other’s opinions as well.

4.2 Results

Based on the results of the discussion session, we conducted a thematic analysis for both experimental groups. For the result analysis, we recorded the gameplay screen of the participants and also the voice of the interview process. Prior notification of how to collect all workshop data was given to all participants before participating in the workshop and their consent was confirmed.

4.2.1 Musically-unskilled Group.

Easy Automatic Composition Experience. The core objective of adapting the Vocal Query Transcription(VQT) model into Studio Mode was to lower entry barriers toward music composition. Our target population reported this system with positive responses.

“A song was written within a few minutes. I never thought that I could compose a music by myself. I might keep playing this game in the future.” (P1, male)

“I think it is very difficult to translate the melody into the sheets, but (in the prototype) it was easy and simple to see the melody changing automatically that I just sang.” (P2, female)

Various Instruments Selection. During the discussion, we have found that participants always wanted to learn or try playing different kinds of instruments, but they couldn’t afford those experiences easily. Thus, hands-on experience with instrumental sounds that participants couldn’t attempt before in real life was mentioned as an satisfying experience.

“I think it’s very interesting to think that the atmosphere of the melody I sang and the genre of the song can change later in the process by selecting and applying various instruments that I have never heard of.” (P1, male)

Intuitive and Specific Feedback. The dialogue and specific feedback when passing or failing in the Classroom Mode were very helpful for the musically-unskilled players.

“Every feedback I got from Professor Bean after each classroom exercise was very accommodating and impactful. It gave me direction for ‘how to’ and ‘what to’ in the next phase.” (P2, female)

4.2.2 Musically-skilled Group.

Simple and Efficient Melody Sketch. Unlike prior experiences in DAW where melody sketches in their brain took complex procedures to be actually represented in the software, Studio Mode reduced all of its inconvenience. Some also noted that they could use this game as a voice memo outside their studio, where they could not record their melody fragments immediately.

“I repeat the process of recording and listening to the notes that I don’t think of when I actually compose, but there is no such thing, so I think it will lower the barrier and be efficient.” (P3, male)

“I want to record it when I suddenly think of something when I compose, and the process of sketching it in front of me was very cumbersome, but (this prototype) seems to simplify this process.” (P5, female)

Unobtainable Experience from other Music Composition Software. Overall, participants who have prior experiences in DAW evaluated the VQT model adapted in Studio Mode as a novel experience. All of the experts agreed and sympathized in using VQT model as a good catalyst for designing a game with composition learning.

“It was fun to compose songs freely with my voice that I didn’t see in the DAW or other MIDI programs I’ve used.” (P4, female)

Proactively Presenting Ideas as Experts. All participants in the group actively presented various ideas that they thought could be helpful for beginners in music composition. Organizing these ideas was very important for our game design since they shared their personal experiences of what was specifically difficult when they were beginners.

“I hope there are tasks in the Classroom Mode where you can freely use Studio Mode’s features before composing.” (P3, male)

“When recording the Beat Layer in the Studio Mode, it would be nice to record the velocity of the beat as well.” (P4, female)

“I wish I could use a keyboard for slightly more advanced management than touching or mouse clicking.” (P5, female)

5 CRITICAL REFLECTION

We started designing *Bean Academy* by questioning ourselves that we listen to music easily every day, but we never get the chance to make our own music. Music composition is known to have very high entry barriers for active use since it requires music theory comprehension, literacy and proficiency in utilizing music composition software, Digital Audio Workstation(DAW) [4, 8, 10]. We defined our main challenges to solve these restrictions on prior requisites when a musically-unskilled population is attempting to learn music composition. As a solution, *Bean Academy*’s Studio Mode was designed with the adaptation of the auditory model-based Vocal Query Transcription(VQT) technique, presenting musically-unskilled players with automatic music generation by just singing and humming short melodies [7, 12]. VQT is being used widely in many other contexts in music search engines, puzzle game mechanisms, vocal training, etc [9, 13]. However, we have not found any prior applied cases to music composition learning in research fields and also the current mobile and PC game market. Based on our playtest, although *Bean Academy* does not outperform the current-level DAW in terms of performance or functionality, it can be highly considered as suitable learning material for musically-unskilled learners. However, we have found that the contents in Classroom Mode were insufficient to support the player’s engagement with Studio Mode. In addition to current contents aimed at boosting the sense of music composition components, we noticed the need to provide alternative guidance and exercises on how to actively use the provided functions in Studio Mode directly.

6 LIMITATION AND FUTURE WORK

In the playtest, participants in both experimental groups mentioned the need for improvements in convenience and accessibility in Studio Mode. More player-friendly functionalities such as cue signs at the beginning of the voice record or no restrictions on music length should be implemented in the future. In addition, working on the debugging process of this prototype version, especially in the editing and deleting MIDI notes in Melody and Beat sublayers, will elicit a better gameplay experience. Not only the enhancements in the main mode, Studio Mode, but our two supplementary modes also need an advancement. Classroom Mode should include more class databases and also the tutorial for Studio Mode as mentioned in the previous section, ‘Critical Reflection’. Club Mode should lead players to continue playing the game by co-composing music with

other players or participating in random mission events for collaborative engagements.

Based on our game, various research can be conducted in the future. First, in our playtest, the ideas we gathered from musically-skilled participants were exceedingly significant. They forgot that they were playtesting participants, and were very active in sharing their past experiences and making improvements for novice learners. Recently, the participatory game design strategy, in which prospective game players or experts participate and design with game designers concurrently, is frequently used in the serious game design and indie game market. As such, for further advancements in our game mechanisms, implementing a game design workshop with musically-skilled experts is expected to bring more effective and valuable results for our game design. Second, by grasping the actual musical sense and composition tendency of musically-unskilled players through music composition data accumulated in the Studio Mode, additional auxiliary learning tools can be designed to compensate for specific deficiencies identified in the data. Finally, with the long-term gameplay of *Bean Academy*, the interest in music composition before and after gameplay or the changes in perception of continuous learning intention can be confirmed through quantitative methods in the future.

7 CONCLUSION

Bean Academy was designed for musically-unskilled learners to lower entry barriers such as music comprehension skills and literacy and proficiency in utilizing the composition computer software, Digital Audio Workstation(DAW). We have implemented a simple and efficient method for music composition learning by adapting the auditory Vocal Query Transcription(VQT) model. Designing the composing gaming experience in the Studio Mode with VQT was aimed to enhance musically-unskilled learners’ satisfaction and enjoyment towards music composition. Through playing three different game modes all-around, players can have a magical experience where their voices suddenly become a piece of music. Overall in the playtest, our target audience reported that *Bean Academy* provided easy-and-simple instructions, intuitive feedback and mostly enthusiastic experience they have never attempted before. From a long-term perspective, we hope that *Bean Academy* will make the musically-unskilled population more accessible to music composition, suppress their vague reluctance to music composition and overall make them think as more satisfying and enjoyable experience.

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